Highlights

UAV-TAlign-12K: Benchmarking Scene-Level Reliability in Cross-FOV UAV Infrared-Visible Alignment

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- UAV-TAlign-12K offers 6,039 synchronized cross-FOV RGB-infrared pairs.
- Seven baselines reveal broadly available pairwise homography evidence.
- UAV-TAlign yields 15 scene transforms and nine high-confidence products.
- Resampling and controlled perturbations validate the operating protocol.

UAV-TAlign-12K: Benchmarking Scene-Level Reliability in Cross-FOV UAV Infrared-Visible Alignment

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Abstract

Cross-field-of-view infrared-visible alignment is a core capability for UAV inspection, nighttime sensing, and multimodal scene understanding. We introduce UAV-TAlign-12K, a benchmark-scale collection of 6,039 synchronized RGB-infrared pairs acquired across 15 industrial and natural scenes, together with an integrity-checked 6,037-pair evaluation split. The benchmark organizes evaluation at three complementary levels: pairwise homography evidence, scene-scale consensus, and reliability–coverage operating analysis. Seven representative methods span classical keypoint pipelines, infrared-visible-specific matching, and modern dense or learned matchers. Pairwise evidence is broadly available: RIFT2, RoMa, and raw MINIMA produce homographies for all 6,037 evaluated pairs, while LoFTR and XoFTR exceed 99% availability. Building on this evidence, UAV-TAlign combines progressive evidence scheduling, geometry-calibrated frame weighting, robust scene consensus, and cross-modal quality verification. It estimates transforms for all 15 scenes and identifies nine high-confidence alignment products at the canonical operating point, covering 74.4% of the evaluated pairs. Five-fold leave-frame-out analysis yields a 1.39-pixel median scene-consensus drift, and controlled translation, rotation, and scale perturbations produce monotonic degradation of the joint quality signal. UAV-TAlign-12K establishes a reproducible foundation for benchmark-scale, confidence-aware UAV infrared-visible alignment.

Keywords: infrared-visible dataset, UAV imagery, multimodal registration, alignment benchmark, scene-level reliability

1. Introduction

Infrared-visible alignment is a foundational capability for UAV inspection, nighttime monitoring, and multimodal thermal sensing. Visible imagery provides semantic and structural detail, while infrared imagery preserves complementary target information across illumination changes, smoke, haze, and low-light conditions. This combination supports multispectral detection, tracking, and aerial perception across demanding operating environments [1, 2, 3, 4, 5, 6, 7, 8]. Accurate cross-sensor geometry is therefore essential for treating the two streams as one sensing system rather than independent observations.

UAV acquisition introduces a distinct alignment problem. Cross-FOV optics, unequal sensor resolutions, changing viewpoints, and zoom variation reshape the same scene differently in the visible and infrared channels. Thermal rendering and day-to-night appearance changes further alter local texture and contrast. A flight sequence consequently provides heterogeneous geometric evidence: individual frame pairs may differ substantially in spatial support, homography stability, and cross-modal agreement. Effective alignment must consolidate this evidence into a scene transform that can support downstream sensing.

Modern correspondence estimation has substantially increased the availability of pairwise evidence. Transformer, dense, and multimodal matchers such as LoFTR, LightGlue, DKM, RoMa, XoFTR, and MINIMA recover correspondences across increasingly difficult appearance changes [9, 10, 11, 12, 13, 14, 15, 16]. Recent UAV alignment benchmarks have also introduced synthetic transform supervision and direct geometric evaluation [17]. These advances establish a strong pairwise foundation. They also motivate a broader benchmark question: how should variable-quality frame estimates be consolidated and assessed at the scale of a captured UAV scene?

We address this question with UAV-TAlign-12K, a real captured cross-FOV UAV infrared-visible benchmark comprising 6,039 synchronized pairs across 15 industrial and natural scenes. Its official evaluation split contains 6,037 integrity-checked pairs spanning day, night, and low-light capture; grayscale and pseudocolor thermal rendering; and wide, zoom, and standard UAV views. The benchmark defines three complementary tracks. Pairwise evaluation measures homography evidence availability, scene-level evaluation measures multi-frame consensus products, and a reliability–coverage profile characterizes selectable operating regimes. This hierarchy connects correspondence generation to the scene products required by UAV sensing systems.

UAV-TAlign serves as the benchmark reference framework. It operates above a strong pairwise matcher and combines progressive evidence scheduling, geometry-calibrated frame weighting, robust scene consensus, and cross-modal quality verification. Across the official split, infrared-specific and modern matchers provide highly available pairwise homographies. UAV-TAlign converts this evidence into transforms for all 15 scenes and identifies nine high-confidence scene products at the canonical operating point, covering 74.4% of the evaluated pairs. Controlled perturbations and leave-frame-out resampling independently support the resulting operating profile.

Our contributions are threefold:

- We introduce UAV-TAlign-12K, a captured benchmark with 6,039 synchronized UAV RGB-infrared pairs across 15 scenes and an integrity-checked 6,037-pair evaluation split.

- We formulate a multi-level protocol spanning pairwise homography evidence, scene-scale consensus, and reliability–coverage analysis, supported by condition-aware diagnostics, resampling stability, and controlled perturbations.

- We present UAV-TAlign, a reference framework that converts strong pairwise evidence into high-confidence scene products through progressive evidence scheduling, robust consensus, and cross-modal quality verification.

2. Related Work

2.1. Cross-Modal Correspondence and Geometric Estimation

Classical registration combines local feature extraction, descriptor matching, and robust model estimation. Scale-space, binary, and nonlinear diffusion features provide efficient geometric references across diverse imaging conditions [18, 19, 20, 21, 22, 23]. RANSAC and its accuracy-oriented successors remain central to converting noisy correspondences into geometric models [24, 25, 26, 27]. For infrared-visible imagery, radiation-variation-insensitive constructions such as RIFT improve descriptor repeatability across modality-specific appearance changes [28]. These methods provide an interpretable foundation for separating correspondence quality from downstream scene aggregation.

Learned correspondence methods have expanded both the robustness and spatial density of pairwise evidence. Early trainable detectors and descriptors established data-driven local representations [29, 30, 31, 32, 33, 34, 35, 36, 37]. Graph reasoning, transformers, and dense warp prediction subsequently enabled direct matching across low texture, repeated structure, and large appearance changes [9, 38, 39, 40, 10, 11, 12, 13, 14, 41, 42]. XoFTR and MINIMA further target visible-thermal or broadly multimodal correspondence [15, 16]. UAV-TAlign builds on this progress by treating pairwise matchers as evidence generators within a scene-scale geometric pipeline.

2.2. Infrared-Visible Alignment

Infrared-visible registration spans feature-based matching, direct optimization, and learned alignment. Radiation-insensitive descriptors address nonlinear cross-modal appearance, while learned formulations estimate pairwise transforms or dense warps from training data [28, 43, 44]. UAV dual-sensor imagery adds cross-resolution optics, viewpoint displacement, thermal rendering variation, and illumination changes. Registration under these conditions has motivated dedicated pairwise methods and UAV-specific data resources.

UAVMatch provides synthetic transform supervision and controlled geometric metrics for trainable UAV alignment [17]. ATR-UMMIM contributes 7,969 real UAV visible–infrared–registered-visible triplets with condition attributes and a semi-automated pixel-level reference [45]. UAV-TIRVis provides 80 manually registered aerial thermal-visible pairs for image-domain evaluation [46]. UAV-TAlign-12K complements these resources with a 6,039-pair captured cross-FOV collection and a multi-frame protocol that connects pairwise homographies, scene consensus, and reliability–coverage operating analysis.

2.3. UAV Multimodal Benchmarks and Scene-Scale Evaluation

Multimodal datasets such as KAIST, FLIR, LLVIP, M3FD, DroneVehicle, VTUAV, LasHeR, and Anti-UAV have advanced detection, fusion, tracking, and low-light perception [1, 2, 3, 4, 5, 6, 7, 8]. Their success demonstrates the value of synchronized visible and infrared sensing. Alignment, however, requires an explicit geometric product and benefits from evaluation beyond isolated correspondence counts.

Table 1: Positioning of UAV-TAlign-12K relative to representative infrared-visible and UAV multimodal resources. UAVMatch denotes the synthetic transform-supervised UAV alignment benchmark.

Resource	Data basis	Alignment signal	Scale	Primary evaluation focus
KAIST / FLIR / LLVIP / M3FD	Ground or mixed RGB-infrared data	Paired or aligned imagery	Dataset-specific	Detection, fusion, low-light perception
DroneVehicle / VTUAV / LasHeR / Anti-UAV	UAV or tracking-centric RGB-infrared data	Task annotations	Dataset-specific	Detection or tracking under multimodal sensing
UAVMatch / TGRS [17]	Synthetic transforms from UAV multimodal data	Affine / homography supervision	81K train / 15K test pairs	Supervised transform regression and geometric metrics
ATR-UMMIM [45]	Real UAV visible / infrared / registered-visible triplets	Semi-automated pixel-level reference	7,969 triplets	Pairwise registration accuracy and condition attributes
UAV-TIRVis [46]	Real UAV thermal-visible imagery	Manual pairwise registration	80 pairs	Direct image-domain registration quality
UAV-TAlign-12K	Real captured cross-FOV UAV infrared-visible data	Multi-frame scene protocol	6,039 candidate / 6,037 evaluated pairs	Pairwise evidence, scene consensus, and operating coverage

UAV-TAlign-12K defines pairwise, scene, and operating-profile tracks. The pairwise track measures the availability and support of homography evidence. The scene track measures how multiple frame estimates form a consensus transform. The operating profile then orders scene products using fixed, interpretable cross-modal diagnostics. This design makes scene-scale evidence consolidation directly comparable across classical, infrared-specific, and modern learned matchers while preserving a common captured-data protocol [47, 48].

3. UAV-TAlign-12K and Evaluation Protocol

3.1. Dataset Scope

UAV-TAlign-12K is a benchmark for real captured UAV infrared-visible alignment. The collection contains 6,039 synchronized RGB-infrared pairs, corresponding to 12,078 images across 15 scenes. Its official evaluation split contains 6,037 integrity-checked pairs and 12,074 images. The collection spans cross-FOV and cross-resolution sensing under day, night, and low-light capture, grayscale and pseudocolor thermal rendering, and wide, zoom, and standard UAV views.

All journal-scale experiments use the same versioned manifest. UAV-TAlign-1K-Lite is retained as a fixed development and ablation subset, while UAV-TAlign-12K defines the benchmark-scale evaluation in this study. The official split comprises 7 day, 5 night, and 3 low-light scenes, together with 8 grayscale-thermal and 7 pseudocolor-thermal scenes. Paired wide and zoom views provide controlled cross-FOV comparisons within the substation scene family.

3.2. Acquisition and Curation

UAV-TAlign-12K was acquired between the second half of 2024 and the first half of 2026 across industrial and natural scenes in southern China. All scenes were captured with the same DJI Matrice 350 RTK platform and

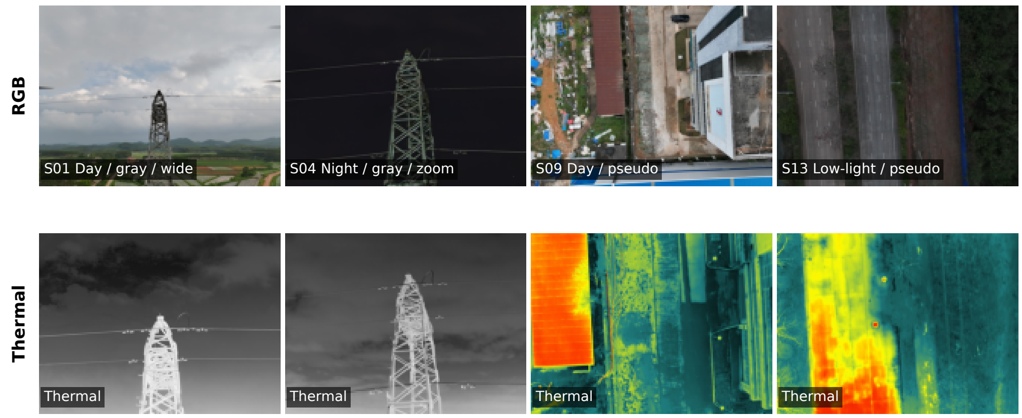

Figure 1: Representative UAV-TAlign-12K RGB-infrared pairs spanning illumination, thermal rendering, and view conditions.

Zenmuse H30T multisensor payload. The integrated payload automatically synchronized the visible and infrared products, preserving the native cross-sensor geometry. The flights combined hovering and route-based capture using a registered aircraft, a CAAC-qualified pilot, and permitted flight areas.

The released streams retain the native H30T acquisition modes. Wide-camera stills are 4032×3024, zoom-camera stills are 3664×2748, and thermal images are 1280×1024. In scenes 01 and 02, 3840×2160 still images were triggered while H30T video recording was active. Scene 07 is a synchronized video-derived sequence sampled at 2 frames per second over one complete orbit around a transmission tower. Grayscale scenes use the H30T *White Hot* setting, and pseudocolor scenes use the on-device *Hot Iron* palette.

Curation criteria were fixed before algorithm evaluation and covered scene relevance, exposure, focus, and file integrity. The resulting candidate collection contains 6,039 synchronized pairs; the integrity audit defines the 6,037-pair official evaluation split. Selection did not use matcher outputs, alignment estimates, or quality scores.

Release preparation removes embedded metadata and normalizes filenames while preserving image pixels. RGB and infrared files are linked through their shared H30T capture key, ordered by capture time and sequence index, and assigned a common six-digit pair identifier. The release applies no resizing, cropping, rotation, recompression, or image enhancement. Guangxi University owns the collection, which is prepared for release under the Creative Commons Attribution 4.0 International license.

3.3. Dataset Organization

Each scene has dedicated `rgb/` and `thermal/` directories, and paired samples share the same filename stem. The official manifest records scene identity, pair identity, illumination, thermal rendering, view type, and scene label. A scene groups frames sharing a site or target context and acquisition condition. It may represent a continuous sequence or multiple sorties under the same scene definition. Scene 07 is the documented single-orbit video sequence.

One manifest supports all pairwise, scene, and condition analyses. It also maps every reported result to its source scene and pair identifier.

3.4. Multi-Level Evaluation Protocol

The benchmark defines three complementary tracks. Track A evaluates each of the 6,037 pairs independently and measures the availability and support of pairwise homography evidence. Track B aggregates frame estimates into a scene transform and reports the canonical set of high-confidence scene products. Track C orders scenes with a fixed, interpretable reliability score to construct coverage-aware operating profiles.

Together, the tracks connect matcher output to deployable scene geometry. Pairwise measurements characterize evidence generation, scene-level measurements characterize multi-frame consensus, and the operating profile exposes how product coverage changes with scene quality.

3.5. Label-Free Scene-Consistency Diagnostics

The protocol uses label-free relative signals for scalable scene-consistency analysis. The stored diagnostics compare the optimized transform with its baseline initialization and quantify changes in cross-modal agreement. They include edge-overlap improvement, gradient-NCC improvement, the fraction of frames with improved gradient agreement, severe baseline-relative outlier count and ratio, accepted-frame support, homography dispersion, and robust-rejection ratio.

These signals jointly characterize cross-modal improvement, geometric support, and consensus stability. The resulting diagnostics support canonical product selection, condition-aware analysis, and reliability–coverage visualization under a common evaluation interface [47, 48].

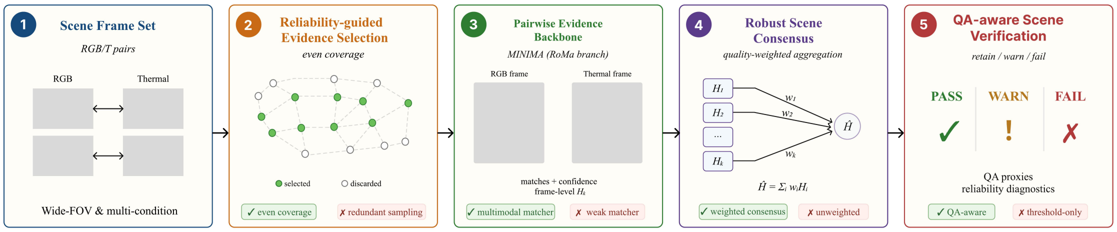

Figure 2: UAV-TAlign reference framework. Progressive evidence scheduling obtains frame-level homographies from a MINIMA-RoMa backbone. Geometry-calibrated weights, median/MAD consensus, and cross-modal verification convert the frame hypotheses into a canonical scene product.

4. UAV-TAlign Reference Framework

4.1. Problem Formulation

UAV-TAlign instantiates the scene-level track of UAV-TAlign-12K. For each scene, the input is a set of paired RGB and infrared UAV frames $S = \{(I_i^R, I_i^T)\}_{i=1}^N$, where viewpoint variation, cross-resolution sensing, thermal appearance changes, and illumination variation produce frame estimates with heterogeneous support. The goal is to recover a thermal-to-RGB scene homography and characterize its operating confidence.

This formulation casts registration as multi-frame evidence consolidation rather than single-pair selection. The framework converts uneven frame estimates into a geometrically stable and cross-modally consistent scene transform. For a selected frame pair, the thermal-to-RGB evidence is represented as a homography $H_i \in \mathbb{R}^{3\times3}$ acting on homogeneous image coordinates,

$$\tilde{\mathbf{p}}_{ij}^R \sim H_i \tilde{\mathbf{p}}_{ij}^T, \qquad H_i = \arg\min_H \sum_{j\in\mathcal{I}_i} \rho\left(\left\|\pi(H\tilde{\mathbf{p}}_{ij}^T) - \mathbf{p}_{ij}^R\right\|_2^2\right), \tag{1}$$

where $\mathcal{I}_i$ denotes the inlier set selected by robust fitting, $\pi(\cdot)$ converts a homogeneous coordinate to image coordinates, and $\rho(\cdot)$ is the robust estimator loss. The scene-level problem is then to estimate a consensus transform $\bar{H}_S$ from the evidence set $\{H_i\}$ and assign its canonical product status.

4.2. Pairwise Evidence Generator

UAV-TAlign uses a pretrained multimodal matcher to generate pairwise evidence. The reference implementation adopts MINIMA with the RoMa branch, while the remaining stages accept homographies and geometry diagnostics from any compatible matcher [14, 16].

4.3. Progressive Evidence Scheduling

UAV-TAlign starts from a compact, evenly distributed frame subset and expands the support set in deterministic stages. The schedule scales with scene size and can cover the complete sequence. Expansion stops when accepted support, consensus stability, and cross-modal quality conditions are satisfied; otherwise, the next subset is evaluated.

Within each stage, evenly distributed sampling provides reproducible temporal or index coverage. The random-selection variant is evaluated separately through a multi-seed sensitivity analysis.

4.4. Geometry-Calibrated Frame Reliability

For each selected pair, the matcher produces cross-modal correspondences and a frame-level homography estimate. UAV-TAlign measures match count, inlier support, inlier ratio, reprojection error, and spatial coverage. A frame enters the scene candidate pool when these statistics satisfy the geometry-quality criterion. This treatment follows established homography estimation and robust fitting practice [25, 24, 26, 27].

Each accepted frame also receives a continuous quality weight. The weight combines inlier ratio, spatial coverage, reprojection quality, matcher confidence, and inlier support, giving greater influence to homographies with stronger geometry. In normalized form, the frame quality weight is written as

$$q_i = \alpha_r\phi(r_i)+\alpha_c\phi(c_i)+\alpha_e\phi(e_i)+\alpha_m\phi(m_i)+\alpha_n\phi(n_i), \qquad \sum_k \alpha_k = 1,\ \alpha_k \geq 0, \tag{2}$$

where r_i is inlier ratio, c_i is spatial coverage, e_i is reprojection quality, m_i is matcher confidence, n_i is inlier support, and $\phi(\cdot)$ maps each diagnostic into a consistent quality direction. The weights remain fixed across all UAV-TAlign-12K evaluations.

4.5. Robust Scene Consensus

Once enough frame-level homographies have been accepted, UAV-TAlign forms a robust scene consensus. Homographies are mapped into parameter space, centered with a median reference, and filtered by a median-absolute-deviation rule to suppress unstable estimates before weighted averaging. The retained homographies are then combined using the frame quality scores defined above. Let $\mathbf{h}_i$ be the scale-normalized vector form of H_i. The consensus set is

$$
\begin{aligned}
\mathbf{h}_S^{\text{med}} &= \operatorname{median}_{j \in \mathcal{A}_S}(\mathbf{h}_j), \\
d_i &= \left\| \mathbf{h}_i - \mathbf{h}_S^{\text{med}} \right\|_2, \\
m_d &= \operatorname{median}_{j \in \mathcal{A}_S}(d_j), \\
s_d &= \operatorname{median}_{j \in \mathcal{A}_S} |d_j - m_d|, \\
\mathcal{C}_S &= \{ i \in \mathcal{A}_S : d_i \le m_d + 2.5 s_d \}.
\end{aligned} \tag{3}
$$

where $\mathcal{A}_S$ is the accepted-frame set before robust consensus. The scene transform is then obtained by quality-weighted aggregation,

$$
\bar{\mathbf{h}}_S = \frac{\sum_{i \in \mathcal{C}_S} q_i \mathbf{h}_i}{\sum_{i \in \mathcal{C}_S} q_i}, \qquad \bar{H}_S = \operatorname{mat}(\bar{\mathbf{h}}_S). \tag{4}
$$

The pipeline records stability diagnostics around the aggregate transform, including homography dispersion relative to the aggregate and the ratio of robustly rejected frame-level estimates. These diagnostics connect consensus formation with the final scene-product decision.

4.6. Cross-Modal Scene Verification

The final stage integrates geometric support with cross-modal scene consistency. UAV-TAlign evaluates the optimized transform on representative frames using edge-overlap improvement, gradient-NCC improvement, the fraction of improved frames, and severe relative outliers. It also records accepted-frame support, mean inlier ratio, mean reprojection error, spatial coverage, dispersion around the aggregate transform, and robust-rejection ratio.

The canonical decision uses a baseline-aware verification rule. High-confidence products combine strong accepted-frame support, stable consensus geometry, controlled baseline-relative outliers, and improved gradient alignment. A fixed, interpretable reliability score summarizes the same diagnostics for operating-profile visualization,

$$
\begin{aligned}
R(S) = {} & \beta_a A_S + \beta_o (1 - O_S) + \beta_j (1 - J_S) \\
& + \beta_e \Delta E_S + \beta_g \Delta G_S + \beta_\gamma \Gamma_S, \qquad \sum_k \beta_k = 1.
\end{aligned} \tag{5}
$$

where A_S is accepted-frame support, O_S is the severe baseline-relative outlier ratio, J_S is the robust-consensus rejection ratio, ΔE_S and ΔG_S are edge and gradient improvement proxies, and Γ_S collects geometry diagnostics. The canonical decision requires a minimum number of accepted frame hypotheses and one of two fixed verification paths,

$$y_S = \mathbb{1}[N_S \geq N_{\min}]\,\mathbb{1}[L_S \vee W_S]\,. \tag{6}$$

Here L_S denotes the primary edge/gradient path, and W_S denotes a high-modality path combining accepted support, consensus geometry, relative outlier control, and gradient improvement. The score $R(S)$ orders scenes for reliability–coverage analysis, while y_S defines the canonical set of high-confidence alignment products.

5. Experimental Setup

5.1. Baseline Suite

We evaluate seven representative baselines using public pretrained configurations without UAV-TAlign-12K-specific adaptation. The suite covers three method families. SIFT and AKAZE provide classical keypoint references, each followed by robust homography estimation. RIFT2 provides an infrared-visible-specific classical reference through its Python implementation. The modern suite contains Kornia LoFTR, RoMa outdoor, XoFTR-640, and raw MINIMA [10, 14, 15, 16].

Raw MINIMA uses the `minima_roma` checkpoint family with the full RoMa branch. RoMa uses the public outdoor full model, and XoFTR uses its 640-resolution visible-thermal checkpoint. LoFTR uses Kornia's public outdoor-pretrained model. Unified inference wrappers standardize image I/O, model output, homography fitting, and result schemas while preserving each pretrained model.

5.2. Geometric Fitting and Deployment Profiles

The shared evaluator converts predicted correspondences into homographies using USAC-MAGSAC, a 3.0-pixel reprojection threshold, 0.999 confidence, 10,000 maximum iterations, and a 4×4 spatial-coverage grid. This common estimator supports matched outcome reporting across the integrated pairwise baselines.

RIFT2 uses its frozen resize-aware profile: `max_dim=1200`, `npt=2000`, `patch_size=64`, Lowe ratio 0.95, and USAC-MAGSAC with a 5.0-pixel reprojection threshold. LoFTR uses resize-aware inference with a longest image dimension of 1200 and automatic mixed precision. Runtime and fitting diagnostics are reported for these declared deployment profiles.

5.3. Protocol Validation

Two controlled experiments evaluate the scene protocol on frozen formal outputs. Five-fold leave-frame-out analysis partitions accepted frame hypotheses deterministically, withholds one fold, recomputes scene consensus, and measures grid displacement from the full-evidence transform. A controlled sensitivity test applies translation, rotation, and scale perturbations at nominal severities of 2, 5, 10, 20, and 40 pixels to three representative frames per scene. The test records edge-overlap and gradient-NCC responses while keeping the canonical decision fixed.

Journal-scale experiments were conducted in isolated Python environments on an NVIDIA RTX 4090 workstation. The main environment used Python 3.10, PyTorch 2.0.1, torchvision 0.15.2, CUDA 11.8, OpenCV 4.11.0, and Kornia 0.6.11. Every run used the official manifest, fixed model weights or interfaces, isolated output roots, and read-only raw dataset directories.

6. Results

6.1. Pairwise Infrared-Visible Homography Evidence

Pairwise infrared-visible homography evidence is broadly available on UAV-TAlign-12K. Table 2 reports the full 6,037-pair official evaluation split. SIFT and AKAZE establish classical keypoint references with 87.08% and 94.29% homography availability, respectively. RIFT2, raw MINIMA, and RoMa produce homographies for every evaluated pair, while LoFTR and XoFTR exceed 99% availability. These results establish infrared-specific and modern dense matchers as strong evidence generators for scene-scale alignment.

Table 2: Pairwise infrared-visible homography evidence on UAV-TAlign-12K. Homography availability reports successful robust estimation, while the remaining columns characterize correspondence support, spatial extent, fitting residual, and runtime under the declared deployment profiles.

Method	Category	OK / N ↑	H (%) ↑	Inlier Ratio ↑	Coverage ↑	Median Reproj. ↓	Runtime / Pair (s) ↓
SIFT + RANSAC	Classical keypoint	5257/6037	87.08	0.242	0.627	0.000	4.421
AKAZE + RANSAC	Classical keypoint	5692/6037	94.29	0.160	0.730	0.105	1.103
RIFT2 (Python, resize-aware)	Infrared-visible classical	6037/6037	**100.00**	0.086	0.354	0.121	13.980
raw MINIMA	Modern dense / learned	6037/6037	**100.00**	0.319	1.000	1.773	1.346
RoMa outdoor	Modern dense / learned	6037/6037	**100.00**	0.328	0.997	1.741	0.929
Kornia LoFTR outdoor (resize-aware)	Modern dense / learned	5984/6037	99.12	0.074	0.895	0.823	0.169
XoFTR-640	Modern dense / learned	5989/6037	99.20	0.089	0.949	1.732	0.647

The method families exhibit complementary evidence profiles. RIFT2 combines complete homography availability with low fitted reprojection error, whereas raw MINIMA and RoMa provide nearly full spatial coverage. Jointly, availability, support, and coverage provide the geometric context for the scene-level benchmark track.

6.2. Scene-Level Reference Result

UAV-TAlign converts the pairwise evidence into scene transforms for all 15 scenes. At the canonical operating point, it identifies nine high-confidence products, representing 60.0% scene coverage and 74.4% pair coverage. The accepted-frame support within these products reaches 1491/2304 (64.7%).

Table 3: Canonical scene-level operating point for UAV-TAlign on UAV-TAlign-12K. Pair coverage and accepted-frame support are computed over the nine high-confidence scene products.

Method	Eval Unit	H Available ↑	High-conf. Scenes ↑	Scene Coverage (%) ↑	Pair Coverage (%) ↑	Accepted / Attempted	Accepted Ratio (%) ↑
UAV-TAlign	15 scenes	15/15	9/15	60.0	74.4	1491/2304	64.7

6.3. Condition-Aware Operating Profiles

Illumination produces distinct scene-product and accepted-support profiles. Day capture provides the highest scene coverage and accepted-frame ratio. Night capture yields 3/5 high-confidence products, while low-light capture preserves a 61.3% accepted-frame ratio across its three scenes. These complementary measurements separate available frame support from the final scene product.

Table 4: Condition-aware operating profile by illumination. Accepted/attempted aggregates frame support over all scenes in each condition.

Condition	Scenes	High-conf. ↑	Scene Coverage (%) ↑	Accepted / Attempted ↑	Accepted Ratio (%) ↑
Day	7	**5**	**71.4**	**1650/2203**	**74.9**
Night	5	3	60.0	361/890	40.6
Low-light	3	1	33.3	462/754	61.3

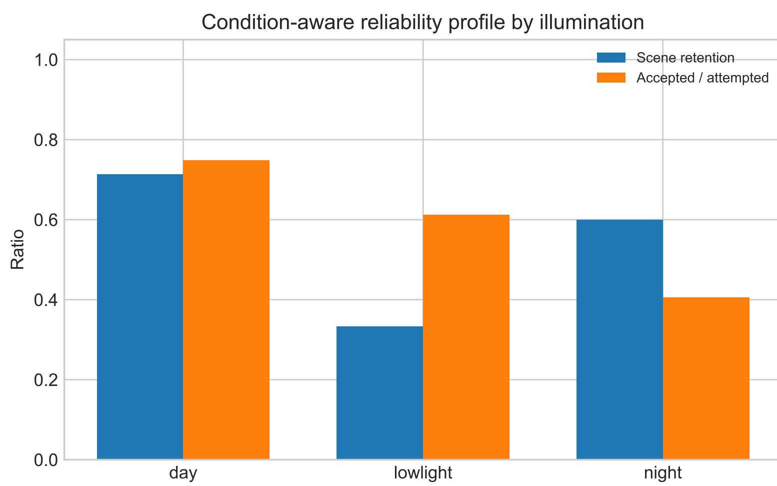

Figure 3: Condition-aware operating profile by illumination. Scene-product coverage and accepted-frame support summarize complementary aspects of scene-scale alignment.

6.4. Reliability-Coverage Operating Profile

The fixed cross-modal verification rule defines the canonical product set. The interpretable score orders scenes for a continuous reliability–coverage sweep. Figure 4 plots this score-ranked profile and overlays the fixed canonical operating point.

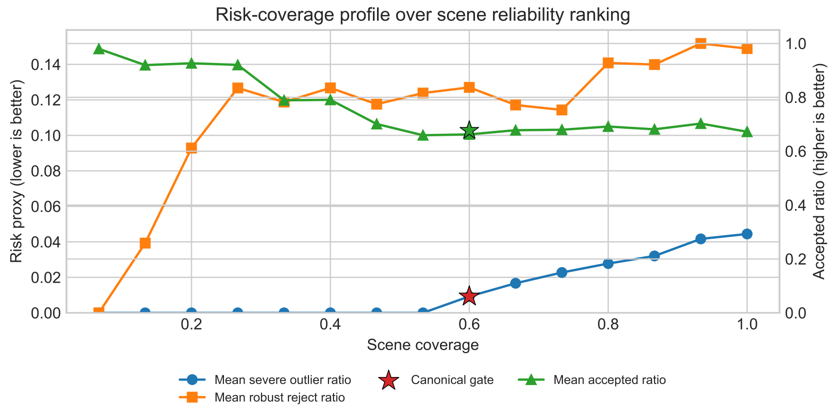

Figure 4: Reliability–coverage operating profile on UAV-TAlign-12K. Curves follow the fixed score-ranked scene order; the star denotes the canonical product set with 9/15 scenes and 74.4% pair coverage.

At the canonical point, UAV-TAlign identifies nine high-confidence products, covers 60.0% of scenes and 74.4% of evaluated pairs, and records a mean severe-outlier ratio of 0.009. The score-ranked curve provides a smooth transition from high-confidence operation to broader coverage.

6.5. Controlled Validation of Scene Diagnostics

The scene diagnostics respond consistently under independent resampling and controlled geometric perturbations. Five-fold leave-frame-out analysis covers all 15 scenes and 75 consensus recomputations. With approximately one fifth of accepted frame hypotheses withheld per fold, the median displacement from the full-evidence consensus is 1.39 pixels (mean 2.26 pixels).

The perturbation test covers 45 representative frames and 675 controlled trials. Within each perturbation family, increasing severity produces a monotonic decrease in the joint edge/gradient quality signal. At the largest severity, translation, rotation, and scale produce mean joint-score changes of -0.074, -0.031, and -0.025, respectively. The corresponding degradation rates are 84.4%, 80.0%, and 73.3%. Table 5 summarizes the independent evidence.

Table 5: Controlled validation of scene-consensus stability and cross-modal quality sensitivity. Synthetic degradation rates and joint-score changes are reported at the largest tested severity.

Validation	Scope	Primary response	Degraded trials (%)
Leave-frame-out consensus	15 scenes / 75 folds	1.39 px scene-median drift	–
Synthetic translation	45 frames / 225 trials	ΔJoint $= -0.074$	84.4
Synthetic rotation	45 frames / 225 trials	ΔJoint $= -0.031$	80.0
Synthetic scale	45 frames / 225 trials	ΔJoint $= -0.025$	73.3

6.6. Cumulative Ablation

The fixed 8-scene ablation isolates robust consensus and cross-modal verification. Robust consensus raises mean edge gain from 0.124 to 0.146 and gradient-NCC gain from 0.126 to 0.149. It simultaneously reduces the severe-outlier ratio from 0.104 to 0.042 and transform dispersion from 151.2 to 52.5 pixels. Cross-modal verification then maps the stage output to the canonical product decision.

Table 6: Cumulative ablation on the fixed 8-scene UAV-TAlign-1K-Lite subset.

Variant	Selection	Consensus	Verification	Stage Pass/N ↑	ΔEdge ↑	ΔGrad ↑	Severe ↓	Disp. (px) ↓
Pairwise evidence only	even	–	–	8/8	0.124	0.126	0.104	151.2
+ Robust scene consensus	even	✓	–	8/8	0.146	0.149	0.042	52.5
+ Cross-modal verification	even	✓	✓	5/8	0.146	0.148	0.042	57.7

Across random-selection seeds, the same fixed subset yields 5/8 to 7/8 scene products while mean accepted-frame support remains comparatively stable. The result confirms deterministic-even selection as a reproducible operating policy and localizes seed sensitivity to a small set of near-threshold scenes.

7. Discussion

7.1. From Pairwise Evidence to Scene-Scale Geometry

UAV-TAlign-12K reveals a transition in the UAV infrared-visible alignment problem. Infrared-specific and modern matchers generate homographies for nearly every pair, providing abundant geometric evidence across the official split. The evidence nevertheless varies in spatial support, cross-modal agreement, and consistency with other frames from the same scene. Scene consensus is therefore a distinct alignment stage: it converts heterogeneous frame estimates into one scene transform that can be used as a common geometric product.

The ablation results identify robust consensus as the dominant module-level contribution. Relative to pairwise evidence alone, consensus improves edge and gradient agreement while sharply reducing severe-outlier incidence and transform dispersion. Cross-modal verification then maps this stable geometry to the canonical high-confidence product set. The combined design links local correspondence evidence with the scene-scale output required by UAV inspection and multimodal sensing.

7.2. Coverage-Aware Operating Analysis

The reliability–coverage profile adds an operating dimension to fixed-threshold evaluation. The canonical point provides nine high-confidence products covering 74.4% of evaluated pairs, while the score-ranked sweep exposes broader coverage regimes. This representation allows an application to select scene-product coverage according to its quality requirements without changing the pairwise matcher or consensus estimator.

Two controlled analyses support the ordering signals used in this profile. Leave-frame-out recomputation preserves the full-evidence consensus with a 1.39-pixel median drift. Translation, rotation, and scale perturbations also produce monotonic degradation of the joint edge/gradient signal. Together, these observations connect the scene diagnostics to both evidence stability and known geometric degradation.

7.3. Benchmark Extensibility

The benchmark separates pairwise correspondence, scene aggregation, and cross-modal verification through a manifest-based interface. Future matchers can therefore be compared as evidence generators, while alternative consensus estimators and operating policies can be studied without changing the captured-data split. The same organization supports extensions to additional scene classes, acquisition geometries, sensor payloads, and downstream infrared sensing tasks.

Deterministic evidence scheduling provides a stable default for this shared interface. The multi-seed study shows that random selection changes the product status of a small set of near-threshold scenes, whereas accepted-frame support remains comparatively stable. This result positions scheduling as a reproducibility mechanism and leaves robust consensus and cross-modal verification as the principal scene-product components.

8. Conclusion

We introduced UAV-TAlign-12K, a benchmark-scale UAV infrared-visible alignment resource with 6,039 synchronized pairs, an integrity-checked 6,037-pair evaluation split, and a multi-level protocol spanning pairwise evidence, scene consensus, and reliability–coverage analysis. Seven representative baselines show that infrared-specific and modern matchers already provide highly available pairwise homographies across captured cross-FOV conditions.

UAV-TAlign converts this evidence into 15 scene transforms. It combines progressive scheduling, geometry-calibrated frame weighting, robust consensus, and cross-modal verification. At the canonical operating point, it identifies nine high-confidence products covering 74.4% of the evaluated pairs. Leave-frame-out resampling and controlled perturbations further support the stability and geometric sensitivity of the scene diagnostics. These results establish scene-scale evidence consolidation and confidence-aware operating analysis as central evaluation dimensions for cross-FOV UAV infrared-visible alignment.

Data Availability

UAV-TAlign-12K, the official evaluation manifest, metadata, integrity report, and evaluation scripts have been prepared for public release. The official evaluation split contains 6,037 integrity-checked RGB-infrared pairs from a 6,039-pair UAV infrared-visible alignment collection. The dataset, owned by Guangxi University, will be released under the Creative Commons Attribution 4.0 International license, with code and evaluation materials provided through a public repository. Review access can be provided during peer review. The official manifest hash is provided for reproducible evaluation.

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